Gain Scheduling Control System for Glucose Level Regulation in Type 1 Diabetes Patients

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Abstract

Type 1 Diabetes (T1D) is a chronic condition in which the body cannot produce insulin, leading to dangerous fluctuations in blood glucose levels. This study uses Simulink to simulate and control glucose-insulin dynamics based on the Hovorka model. A First-Order Plus Dead Time (FOPDT) approximation was obtained for three glycemic scenarios: normoglycemia, hypoglycemia, and hyperglycemia. Based on these models, PID controllers were tuned using the Dahlin method to restore glucose to a normoglycemic range (~5.5 mmol/L). Simulations show that the proposed controllers effectively stabilize glucose levels and reduce deviations. These results support the use of PID-based control as a foundation for future adaptive insulin delivery systems.

1. Introduction

Type 1 Diabetes (T1D) is a metabolic disorder characterized by the inability of the pancreas to produce insulin, a hormone essential for maintaining blood glucose levels within a healthy range. Without proper insulin regulation, patients face risks of acute complications such as hypoglycemia and hyperglycemia, along with long-term damage to organs and systems[2]. Effective control of blood glucose is therefore critical to improving the quality of life and long-term outcomes of individuals living with T1D.

Traditional insulin therapy often relies on manual dosing or fixed-rate insulin pumps, which lack adaptability to real-time physiological changes. This has motivated the development of closed-loop systems capable of automatically adjusting insulin delivery based on continuous glucose monitoring. Among various control strategies, Proportional–Integral–Derivative (PID) controllers stand out for their simplicity, robustness, and successful application in industrial and biomedical systems.

This work is based on the Hovorka model [1], a physiologically validated mathematical representation of glucose-insulin interactions in T1D patients. Simulation environments such as those proposed by Wilinska et al. [2] offer a safe and effective way to test these models. Recent studies have also proposed improvements to the original equations to better reflect clinical data, as shown by Som et al. [3].

The aim of this study is to evaluate the performance of a gain scheduling control strategy based on PID controllers, designed using FOPDT approximations of the Hovorka model across different glycemic conditions. The controllers were tuned using the Dahlin equations and implemented in Simulink to test their ability to regulate blood glucose toward a normoglycemic setpoint.

The rest of this paper is organized as follows: Section 2 provides background on the modeling approach and PID control. Section 3 explains the methodology used for system identification and controller tuning. Section 4 presents the simulation results. Finally, Section 5 summarizes the conclusions and suggests directions for future work.

2. Background

2.1. Glucose-Insulin Model

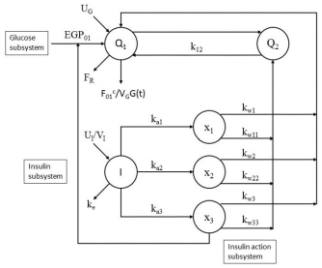


Figure 1. *Glucose-Insulin Model*

The glucose-insulin system maintains blood sugar levels. In Type 1 Diabetes (T1D), this regulation fails, requiring external insulin delivery. The Hovorka model divides the system into glucose dynamics, insulin kinetics, and insulin action to capture key physiological interactions [1].

2.2. Mathematical models of the system

The Hovorka model [1] is used in this work to simulate glucose-insulin regulation. It divides the physiological behavior into three subsystems:

- **Glucose kinetics:** Describes the glucose dynamics in accessible (Eq. 1) and non-accessible (Eq. 2) compartments, as well as non-insulin dependent glucose uptake (Eq. 3), renal clearance (Eq. 4), and glucose absorption from meals (Eq. 5).

$$\frac{dQ_1(t)}{dt} = EGP_0 + U_G + 0.01Q_2 + [x_1k_{w1} + x_2k_{w2} + x_3k_{w3}] - F_RQ_1 - \left[\frac{F_{01}{}^C}{V_G G(t)}\right]Q_1 - 0.002Q_1 \quad (1)$$

$$\frac{dQ_2(t)}{dt} = [k_{w11}x_1(t) + k_{w22}x_2(t) + k_{w33}x_3(t)] + EGP_0[k_{w1}x_1(t) + k_{w2}x_2(t) + k_{w3}x_3(t)] - k_{12}Q_2 \quad (2)$$

$$F_{01}{}^C = \begin{cases} F_{01} & \text{if } G \geq 4.5 \text{ mmolL}^{-1} \\ F_{01}G/4.5 & \text{otherwise} \end{cases} \quad (3)$$

$$F_R = \begin{cases} 0.003(G-9)V_G & \text{if } G \geq 9 \text{ mmolL}^{-1} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

$$U_G = \frac{D_G A_G t e^{-t/t_{max,G}}}{t_{max,G}{}^2} \quad (5)$$

- **Insulin kinetics:** Represents the transport and distribution of insulin in the body using two compartments (Eqs. 6–9).

$$\frac{dS_1(t)}{dt} = u(t) - \frac{S_1(t)}{t_{max,I}} \quad (6)$$

$$\frac{dS_2(t)}{dt} = \frac{S_1(t)}{t_{max,I}} - \frac{S_2(t)}{t_{max,I}} \quad (7)$$

$$\frac{dI(t)}{dt} = \frac{U_I(t)}{V_I} - k_e I(t) - [k_{w1}x_1(t) + k_{w2}x_2(t) + k_{w3}x_3(t)] \quad (8)$$

$$U_I = \frac{S_2(t)}{t_{max,I}} \quad (9)$$

- **Insulin action:** Captures the insulin effects on glucose utilization, uptake, and production via three action compartments (Eqs. 10–12).

$$\left[\frac{dx_1}{dt}\right] = k_{a1}x_1(t) + k_{w1}I(t) + k_{w11}I(t) \quad (10)$$

$$\left[\frac{dx_2}{dt}\right] = k_{a2}x_2(t) + k_{w2}I(t) + k_{w22}I(t) \quad (11)$$

$$\left[\frac{dx_3}{dt}\right] = k_{a3}x_3(t) + k_{w3}I(t) + k_{w33}I(t) \quad (12)$$

2.3. Identification theory

To design a control system for glucose regulation, it is necessary to approximate the behavior of the physiological model with a simpler representation. This process is known as system identification.

In this project, the glucose response to insulin was approximated using a **First-Order Plus Dead Time (FOPDT)** model [5], commonly used in control applications due to its simplicity and ability to approximate the dynamics of systems with a delayed response.

The general transfer function of the FOPDT model is:

$$G(s) = \frac{Ke^{-\theta s}}{\tau s+1} \quad (13)$$

Where K is the process gain, τ is the time constant, θ is the time delay.

The **Step Response Method** was used to estimate these parameters. The time constant τ and the delay θ were extracted from the output signal after a step change in insulin. Specifically, the times at which the response reached 28.3% and 63.2% of its final value were used. The gain K was calculated from the steady-state change.

The following expressions were used for the parameter estimation:

$$K = \frac{\nabla output}{\nabla input} = \frac{glucose_{final} - glucose_{inicial}}{insuline_{final} - insuline_{inicial}} \quad (14)$$

$$\tau = 1.5 \cdot (t2 - t1) \quad (15)$$

$$\theta = t2 - \tau \quad (16)$$

Interpolation was used to determine the exact times t1 and t2 at which the output reached 28.3% and 63.2% of the final value, respectively.

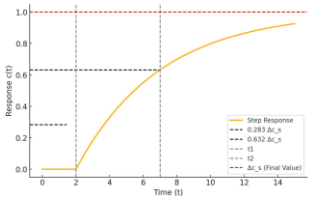


Figure 2. *FOPDT identification*

2.4. Control theory

Gain Scheduling is a control strategy used in nonlinear or time-varying systems, where a set of linear controllers is designed for different operating conditions and switched or interpolated based on measurable variables [5].

The control strategy used in this work is based on **Gain Scheduling**, a technique where multiple controllers are pre-tuned for different operating conditions. In this case, the patient's glucose level is monitored in real time to select the most appropriate PID controller (for normoglycemia, hypoglycemia, or hyperglycemia). Each controller has specific gains adapted to its range, allowing the system to maintain stable glucose regulation even when the patient's state changes.

To implement this strategy a Proportional-Integral-Derivative (PID) controller was implemented to regulate blood glucose levels by adjusting the insulin infusion rate according to the error between the actual glucose value and the desired setpoint.

The PID control law is given by:

$$U(t) = K_c e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \quad (17)$$

Where e(t) is the control error (setpoint – glucose), and Kc , Ki, and Kd are the proportional, integral, and derivative gains, respectively.

2.4.1. PID Tuning

To compute the controller parameters, the **Dahlin tuning method** was applied, using the parameters obtained from the identified FOPDT model for each glycemic condition. This approach provides a good balance between performance and robustness, and is well suited for systems with delays, such as the glucose-insulin dynamics.

The Dahlin tuning formulas are:

$$tc = \frac{1}{5} \times \theta \quad (18)$$

$$td = \frac{\theta}{2} \quad (19)$$

$$kc = \frac{\tau}{K \times (\theta + tc)} \quad (20)$$

$$ti = \tau \quad (21)$$

$$ki = \frac{kc}{\tau} \quad (22)$$

$$kd = kc \times \tau \quad (23)$$

Where kc, ki, kd are the PID gains.

3. Methodology

3.1. Matlab model

The simulations were carried out in MATLAB/Simulink using the Hovorka model, a well-established representation of glucose-insulin dynamics in type 1 diabetes. The model captures the interaction between insulin infusion, glucose absorption, and insulin action, allowing for realistic simulation of the glycemic response under different physiological states.

3.2. Model parameters

The initial conditions of the system represent a basal state without carbohydrate intake and were used in all simulations. The values are summarized in Table 1.

Glucose(mmol/L)	Insuline(mU/L)	UG(mmol/min)
5.72790	10.35	0

Table1. *Initial conditions used in the simulations*

UG refers to exogenous glucose intake.

3.3. Identification

A **First-Order Plus Dead Time (FOPDT)** model was identified to approximate the behavior of the Hovorka model under three different glycemic conditions.

The identification was performed separately for each scenario:

Hypoglycemia: glucose < 4 mmol/L (72 mg/dL)

Normoglycemia: 4 ≤ glucose ≤ 7 mmol/L (72–126 mg/dL)

Hyperglycemia: glucose > 7 mmol/L (126 mg/dL)

For each condition, the parameters K, τ, and Θ were estimated using the step response method, as summarized in Table 2.

Condition	K	τ	Θ
Normoglycemia	-59.2274	284.1014	120.7215
Hypoglycemia	-13.0191	153.7783	86.8692
Hyperglycemia	-82.4816	274.8000	119.6000

Table 2. *Identified FOPDT parameters for each glycemic condition*

3.4. Control

The PID controllers were tuned for each of the three glycemic scenarios identified in Section 3.3.

The PID gains for each condition were computed using the Dahlin tuning method, as detailed in Section 2.4.1. These controllers were then integrated into a Gain Scheduling framework, which dynamically selects the appropriate set of gains based on the patient's glucose level in real time.

This adaptive strategy ensures robust glucose regulation under varying conditions by automatically switching between the tuned controllers for each glycemic state

The resulting PID parameters for each glycemic scenario are summarized in Table 3.

Condition	Kc	Ki	Kd
Normoglycemia	-0.0331	-0.00011655	-9.4072
Hypoglycemia	-0.1133	-0.00073684	-4.9216
Hyperglycemia	-0.0232	-0.00008448	-1.3882

Table 3. *PID controller parameters obtained using the Dahlin method.*

3.5. Software used

All simulations were performed using MATLAB R2023a and Simulink. The models were implemented using standard Simulink blocks and custom subsystems representing the Hovorka equations and PID controller.

3.6. Simulation enviroment

The glucose-insulin dynamics and the PID controllers were implemented in Simulink. Two types of controllers were tested: a fixed PID controller and a variable PID controller with gain scheduling. A gain scheduling control strategy based on PID controllers was implemented using a MATLAB Function block that selects the controller gains based on the current glucose level. The PID structure was then applied to compute the insulin infusion rate. Both implementations are shown in Figures 3 and 4.

To evaluate the performance of the controllers, simulations were carried out by changing the glucose setpoint from the normoglycemic level to hypoglycemic and hyperglycemic levels. This allowed testing the ability of each controller to restore glucose levels under varying conditions.

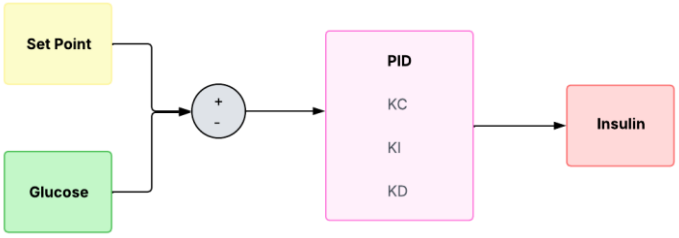


Figure 3*. Fixed PID controller structure.*

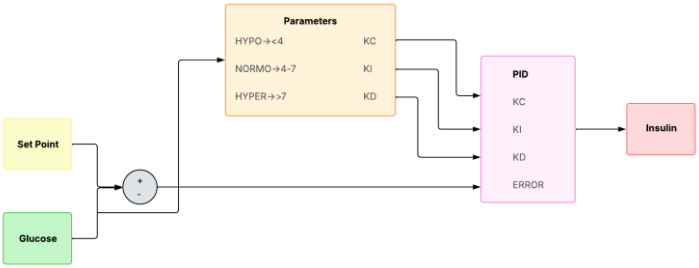


Figure 4*. Gain scheduling strategy based on PID controllers*

To simulate a meal, the glucose input UG was connected in Simulink.

4. Results

4.1. Comparison Between Real and Identified System

The accuracy of the system identification was assessed by comparing the response of the Hovorka model and the corresponding FOPDT model in three glycemic conditions: normoglycemia, hypoglycemia, and hyperglycemia.

As shown in Figures 5 to 7, the FOPDT model closely matches the behavior of the Hovorka model, capturing the main features of the dynamic glucose response to insulin administration, including the delay and the gradual convergence to steady state.

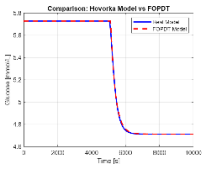


***Figure 5**. Real vs Identified Model Normoglycemia*

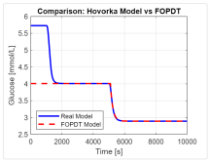


***Figure 6**. Real vs Identified Model Hypoglycemia*

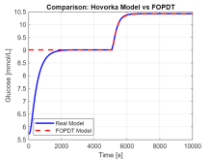


***Figure 7**. Real vs Identified Model Hyperglycemia*

4.2. Control Experiments

Two PID control strategies were tested: a gain-scheduling (variable) PID and a fixed-parameter PID. Both were evaluated under identical experimental conditions, including a simulated carbohydrate intake of approximately 50 mmol/min at 2000 seconds.

The figure below shows the glucose response for both controllers over the entire simulation period:

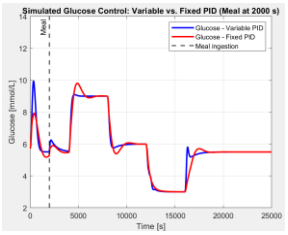


***Figure 8**. Glucose response with fixed and gain-scheduling PID controllers*

The **gain-scheduling PID controller** achieved faster stabilization and adapted more effectively to changing glycemic conditions throughout the simulation. It also showed a shorter settling time compared to the fixed PID controller.

The **fixed PID controller** exhibited slower recovery, higher peaks, and a more prolonged return to the target value.

At the time of the simulated meal (2000 seconds), both controllers responded in a comparable and stable manner, showing no significant difference in glucose regulation during that specific event.

5. Discussion

In this work, two types of strategies were compared for glucose regulation: a fixed-gain PID and a gain-scheduling PID that adapts its parameters to different glycemic conditions.

Figure 8 shows that the gain-scheduling PID controller exhibited a higher initial overshoot compared to the fixed PID controller. However, as the simulation progressed, the overshoot was progressively reduced. The gain-scheduling strategy adapted better to changes in glycemic setpoints, resulting in smoother transitions and improved long-term performance. In contrast, the fixed PID controller maintained a more rigid response and did not adjust as effectively to variations in glucose levels.

Other studies also confirm that PID controllers offer a good balance between simplicity and performance. While advanced methods like Model Predictive Control (MPC) may improve prediction, they are more complex to implement and computationally demanding [4]. In contrast, the PID controller is easier to tune and works well in real time, especially when adapted with gain scheduling [5].

6. Conclusion

This project has shown that using a gain-scheduling PID controller improves glucose regulation in patients with type 1 diabetes, compared to a fixed PID controller. The adaptive controller was able to respond to different glycemic conditions and recover more effectively from disturbances.

The strategy was implemented in Simulink using the Hovorka model and a simplified FOPDT version to support parameter tuning. Simulations demonstrated that this method can keep glucose levels close to the target value (5.5 mmol/L) across all scenarios.

In summary, the gain-scheduling PID controller is a simple, adaptable, and effective approach that could be integrated into closed-loop insulin delivery systems for people living with type 1 diabetes.

Acknowledgements

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